

# B2B I

: v1.1 | : 2026-07-27 | : &

## 1.

B2B .1 , . . .  
(Incremental Cash Margin) .  
 , .

## 2. (Trade-off Matrix)

### 2.1 (Frontend)

| &                                |                         |       |         | (Trade-off)                       |     |           |     |
|----------------------------------|-------------------------|-------|---------|-----------------------------------|-----|-----------|-----|
| React 19 + Vite + TypeScript ( ) | B2B ,                   | SPA,  | HMR     | vs Next.js / API . Vite           | SEO | SSR       | BFF |
| TanStack Query v5 ( )            | API ,                   |       | ,       | vs fetch / .                      |     | API       |     |
| URL Search Params                | (2F~1F), URL            |       | , D-Day | vs In-memory state , URL 100%     |     | URL       |     |
| Zustand ( )                      | UI .                    |       |         | vs React state / selector         |     | UI        |     |
| TanStack Table v8                |                         | SKU , | , ,     | vs Basic Table / AG Grid Headless |     | shadcn/ui |     |
| Recharts                         | vs                      | vs    | ,       | vs Chart.js / D3.js React SVG     |     | D3        |     |
| Tailwind CSS v4 + shadcn/ui ( )  |                         |       | .       | vs MUI /                          |     | .         |     |
| React Hook Form + Zod            |                         |       |         | vs Formik / Plain State od        |     | Zod Z     |     |
| Web Vitals                       | LCP < 1.2s, INP < 100ms |       | UX      | vs                                |     |           |     |
| : Vitest + RTL + Playwright ( )  | .                       | .     | E2E     | vs Jest/Cypress / Vite-React CI   |     | .         |     |

2.2 (Backend)

| &                                      |                             |  |  | (Trade-off)                                   |                   |          |        |
|----------------------------------------|-----------------------------|--|--|-----------------------------------------------|-------------------|----------|--------|
| Java 21 + Spring Boot 3.x ( )          | REST API, . . .             |  |  | vs Node.js/Python / FastAPI                   | OpenAPI-timeout . | Trace ID |        |
| Spring Data JPA + QueryDSL 5.x ( )     | CRUD .                      |  |  | vs Native SQL/jOOQ / QueryDSL, read model-SQL | JPA-SQL           | r        |        |
| MyBatis 3.5+                           | / Function/CTE , Window SQL |  |  | vs JPA SQL XML                                | XML               |          |        |
| (MVP)                                  | → → → → .                   |  |  | vs State Machine / num .                      | E .               | State M  | achine |
| Spring Security OAuth2/OIDC + RBAC ( ) | .                           |  |  | vs JWT / IdP R esource Server                 | IdP .             |          |        |
| Spring Batch 5 + 1                     | .                           |  |  | vs @Scheduled/Quartz/Kubern etes CronJob / .  | Batch Quartz      |          |        |
| JUnit 5 + Mockito + Testcontainers     | Docker PostgreSQL/Redis     |  |  | vs H2 DB H2 PostgreSQL JSONB/SQL              | Docker            |          |        |

2.3 (Database & Cache)

| &                  |                       |  |  | (Trade-off)             |   |   |           |
|--------------------|-----------------------|--|--|-------------------------|---|---|-----------|
| PostgreSQL 16+ ( ) | . . . , payload JSONB |  |  | vs MySQL/Oracle / JSONB | . | . | .         |
| Redis 7.x ( )      | .LLM TTL              |  |  | vs / TTL                | . | 0 | .stampede |

2.4 AI (AI Service)

| &                                |       |  |  | (Trade-off)               |                    |                   |   |
|----------------------------------|-------|--|--|---------------------------|--------------------|-------------------|---|
| Python 3.11+ + FastAPI ( )       | .ML   |  |  | vs Spring AI / Python .ML | HTTP .timeout .    | CPU               |   |
| OR-Tools/HiGHS PuLP + solver ( ) | . . . |  |  | vs / . . MILP             | ‘100% ’ . .        | infeasible solver |   |
| scikit-learn / LightGBM +        | .     |  |  | vs RAG/Vector DB /        | .                  | .                 | . |
| LLM Provider Adapter ( )         | .     |  |  | vs / . . .                | PII .JSON Schema . | fallback .        |   |

2.5 , , Sentry (Observability & Testing)

| & (Trade-off)          |                                         |                        |         |
|------------------------|-----------------------------------------|------------------------|---------|
| Sentry ( )             | .                                       | vs /                   | . .     |
| Prometheus + Grafana   | CPU, , DB , API R PS/Latency            | vs CloudWatch          |         |
| + Grafana Loki ( )     | . .                                     | vs ELK /               | . .PII  |
| OpenTelemetry + Jaeger | React->Spring Boot->FastAPI->PostgreSQL | vs TraceID             | TraceID |
| k6 ( )                 | . . API SLA P95/P99.                    | vs JMeter/Locust / SLO | 500     |

3. AI & 4

1. ML (Feature Engineering) & : , D-Day, 30/60/90 , , LightGBM .
2. / SKU (Cold-Start) : -> -> , 65% .
3. LLM (Structured Output) & : LLM JSON , TTL. .PII .
4. (Closed-Loop Feedback System): ( , ) DB AI (MAPE) , .

4.

Frontend (React SPA) -> Reverse Proxy (Nginx) -> Core Backend (Spring Boot) -> AI Microservice (Python FastAPI) -> Database (PostgreSQL/Redis) -> Observability (Prometheus/Loki/Sentry) .

## 5. 面试问题

- 面试官问：你用过 React 吗？  
答：用过，React 19 + Vite + TypeScript。SEO API 用 Vite 做，  
TanStack Query, Zustand API 用 Spring Boot FastAPI OpenAPI , timeout Trace ID  
JPA/QueryDSL MyBatis . CRUD QueryDSL, read model SQL  
MVP Enum . State Machine  
Spring Batch Quartz Quartz, @Scheduled  
PostgreSQL . payload JSONB . Redis TTL 0  
OR-Tools/HiGHS PuLP solver  
infeasible-solver fallback  
LightGBM , '100%'  
LLM , Provider Adapter JSON Schema PII

### 5.1

- : SKU , , ,
- : IdP/OIDC, RBAC, ,
- : , , , fallback,
- : LLM . DB , timeout .

### 5.2

- P0: React/Vite , Spring API , PostgreSQL ,
- P1: FastAPI , ,
- P2: LLM , State Machine,